

Steel Framing: How Green?

Q: Is the use of steel framing for residential construction turning into a trend that we must now take seriously? Isn't energy efficiency very tricky when using steel? When should an energy-efficient custom builder like me decide to use steel framing?

—Stumped by Steel

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A: Steel framing is made of light-gauge, cold-formed steel. All its structural components—including exterior studs, tracks, columns, rafters, beams, purlins, and fasteners—are manufactured to exacting specifications designed to make steel frame construction a viable alternative to wood frame residential construction.

This construction method is gaining popularity, as is shown by the fact that a growing number of homes use steel construction. In 1995, an estimated 55,000 homes were built with steel framing; by 1999, this figure had jumped to 87,000. Light-gauge steel shipments for residential construction increased 41% from 1998 to 1999. By 2003, the Steel Recycling Institute estimates that 25% of all new homes built in the United States will be totally or partially steel framed. This is a new construction method that cannot be ignored.

In manufacturing, sheet steel made at steel mills is roll-formed into the proper shape for framing members. The resulting pieces are galvanized with zinc to prevent rust and corrosion. In cross section, studs are typically C-shaped and the upper and lower tracks in which they are held are U-shaped, with standard dimensions similar to wood framing. These

framing members are made to thicknesses of 0.018–0.097 inches (18–97 mils).

Advantages of Steel

Steel has many advantages over wood. It won't warp, splinter, or crack. Steel is also pest resistant and will not rot. When installed properly, steel framing effectively resists hurricanes and other types of extreme weather. Steel framing is engineered to strict standards, reducing construction and trim waste, as well as losses from poor lumber pieces. These factors also reduce site cleanup and waste hauling time (and costs).

Steel framing will not burn, so homes that are made with it have a lower fire risk than homes that are not. In at least three states (Indiana, Texas, and Hawaii), some insurance companies offer lower homeowner insurance premiums on steel-framed homes for this reason. And steel framing has also attained the highest earthquake shock standard.

Although it is currently more expensive, one of the big benefits of steel framing is that the price of steel is more stable than that of lumber. Contractors need not plan purchases and delivery of framing members nearly so carefully as they do with wood. Longer price guarantees are available to builders, and project budgeting should be easier.

Steel has the highest strength-to-weight ratio of any common building material. For design, the span strength of steel allows for the option of larger interior spaces for occupants. And contrary to what one might think, steel-framed homes do not exhibit any additional difficulties with respect to television and radio reception.

Installation

When used in residential construction, steel framing members are typically substituted for wood on a one-for-one basis for both load-bearing and nonload-bearing applications. There is one difference to begin with, and that is that steel studs, joists, and rafters fit into top and bottom U-shaped tracks. Additional skill is needed when installing steel framing, as the top track is not capable of transferring vertical loads; the other framing members must be aligned vertically to transfer vertical loads. This n-line framing, as it is called, requires more precision than typical wood construction.

Although basic tools can be used in constructing steel framing, additional experience is useful when cutting members with chop saws, aviation snips, or electric shears. Fastening is more labor intensive than using wood nailers, as the fundamental fasteners for steel framing are



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and warping of typical house framing materials. This also ensures straight walls and square corners, so that drywall can hang straight and will

screws. (Using a screw gun is very different from using a nail gun.) Fastening steel members requires first clamping the assembly and then driving the screw. New fastening systems using pneumatics (as well as crimping and welding) are under development; these methods will speed the construction process.

One of the benefits, however, of using steel in this way is that the framing members are available with prepunched holes that can be used for relatively quick installation of electrical wiring and plumbing. (This is typically done in interior walls, where air sealing isn't a problem.) In addition, drywall installation is easier once drywallers are trained. Steel studs are identical to one another, without defects. The dimensional stability of the studs eliminates nail pops in the drywall from the drying

not bulge from bowed framing members. Contraction and expansion from temperature changes are too small to result in interior drywall cracking.

Current Limitations

Steel framing is not yet standardized, so it's impossible to use products or systems interchangeably. The most important design variables in framing members include the web, the flange, the lip, the bend radius, the thickness, and the yield strength, along with the size, shape, and location of web cutouts (see Figure 1). These can vary by manufacturer, and even a small variation can be important.

Steel framing is incorporated into all major building codes; however, not all local code officials and inspectors are familiar with the use of steel in residential construction. This can lead to complications with code officials, which—along with the extra cost—may discourage builders from using residential steel framing.

Green Steel

Building with steel has green pluses and minuses. These are revealed by examining steel as a material, its energy impact, its safety, its disposal, and its cost.

Materials use. Steel is the most widely recycled material in the United States—about 65% of domestic steel is recycled. This includes *all*

steel, from that used in automobiles, appliances, and industry to that used in food cans. Because of the recycling infrastructure, steel is also the most recyclable material—it is even more recyclable than aluminum. Every ton of steel recycled saves 2,500 lb of iron ore, 1,400 lb of coal, and 120 lb of limestone from being mined.

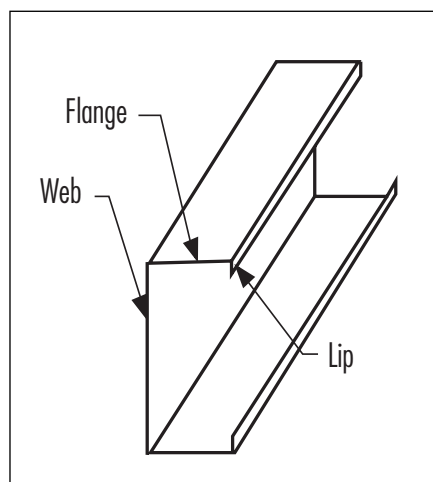
As I mentioned earlier, engineered standards for steel framing help to reduce construction waste. Framing members are available cut to length; this reduces on-site waste.

Steel framing weighs up to two-thirds less than wood. While wood framing for one house requires up to 1 acre of mature trees, steel framing for the same house uses an equivalent of just four to six junked cars.

Energy impact. A significant portion of manufactured steel is recycled steel. The total amount of steel recycled annually saves enough energy to supply electric power to approximately 18 million homes for one year.

Steel conducts heat extremely well—approximately 400 times bet-

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Figure 1. Some steel framing members come with features such as a lip and web cutouts.

ter than wood does. This results in a process called thermal bridging, whereby heat tends to choose the most conductive path. This phenomenon lowers the effectiveness of cavity insulation. It can also sometimes cause black stains on interior house surfaces, at points where there are cold spots in the walls (see “Staining Patterns and Physical Forces,” *HE* Nov/Dec '98, p. 20).

Several methods can be used to attempt to counteract the thermal



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bridging effect of steel-framed walls; each works by interrupting the path along which heat flows. For cold climates, installation of rigid exterior insulation, such as sheathing, is recommended in order to provide a thermal break. Another way to meet the R-value requirements of the Model Energy Code (MEC) is to increase the amount of cavity insulation. Additional attention is needed where steel walls rest on concrete foundations, where roof truss members transfer attic heat to the interior, and at bottom floor joists located over unheated spaces. The North American Steel Framing Alliance (NASFA) publishes thermal guidelines that address these issues.

In addition, studs can be spaced farther apart than standard practices allow without affecting the integrity of the structure, thereby reducing the possible conductive paths of heat in the structure. Finally, punching large holes in the web without affecting the structural integrity of the stud can reduce thermal paths and thus thermal bridging. Thus structural redesign is needed because of the unique properties of steel. (See "Steel Stud Walls: Breaking the Thermal Bridge," p. 39.)

What about total embodied energy, of which recycling

contributes only a portion? A public/private partnership called the Consortium for Advanced Residential Buildings did research on similar 1,448 ft² town houses made from different materials, and concluded that the embodied energy for all materials was insignificant compared to the operational energy required to run the building.

Safety. Unlike wood, steel does not have to be treated with toxic compounds to be protected from pests and decay. Thus, using steel as a structural element can prevent the use of chemically treated lumber. It must be kept in mind, however, that steel manufacture depends strongly on the use of coal—both as a fuel source and as coke in the blast furnace. Coal is one of the most polluting energy sources that is currently used, as it contains a variety of harmful chemicals. Combustion products include sulfur oxides, nitrogen oxides, particulate matter in the form of ash, and radioactive elements such as uranium and thorium.

Disposal. Because steel is recycled at significant rates, waste from framing construction, as well as old framing demolition, is recycled rather than landfilled. On-site waste is minimal, typically about 2%, and

the energy saved by significant recycling is substantial. In many ways, the disposal factor leans strongly in the favor of steel.

Costs. Steel framing is currently priced, on a nationwide average, slightly higher than wood framing (estimated at about a 5% differential in 1998 in a study supported by the U.S. Forest Service). This is due to the labor costs of fastening and adding insulation to reduce thermal bridging. The National Association of Home Builders (NAHB) estimates that steel framing costs are in the range of \$400–\$450/1,000 board ft equivalent. As long as lumber prices do not exceed this range, wood has the cost advantage.

Over time, the price differential enjoyed by wood will probably reverse if additional pressure is put on forest reserves and both economies of scale in, and labor experience with, steel framing production are realized. Ultimately, consumer demand is likely to determine whether steel framing will become a widespread alternative to wood.



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